SIEMENS

TEST CERTIFICATE
FOR
THREE PHASE INDUCTION MOTOR

Test Certificate as per IEC 60034-1: 2010 & IS 12615:2011

Certificate No.	2015 I019 K-2015040	**Date:**	21-Mar-18
Customer Details:		**Test Date:**	24-Feb-16

Details of Machine

Machine Sr No.	K-2015040	Type/Model:	1LE7501-0DB32-3AA4	IE Code	IE2	
Voltage	415 V	Connection	Y	Insulation class		F
Frequency	50 Hz	Direction of Rotation	CW	Utilization		B
Current	1.7 A	Duty	S1	Site Altitude		<1000m
Power	0.75 kW	Torque/ Rotor class	kl16			
Power Factor	0.75	Type of Construction	IMB3	Ambient Temp.(°C)		50
Speed	1433 RPM			Bearings:	DE	6204 ZZC3
Efficiency(η)	80.4 %	Protection	IP55		NDE	6204 ZZ C3
Type	Motor	Tst/Tn	2.8	Frame Size:		80
Manufacturer	Siemens Ltd.	Ist/In	5.8			
Other Data:						

Condition of machine on receipt: **Good**

Test Details

Sr No.	Test Details	Test Specification reference	Performed	Tolerances or Acceptance Criteria		
				Parameter	Criteria	
1	Resistance measurement of Stator winding	IEC 60034-2-1:2007 cl.5.7	Yes	-	-	-
2	Temperature rise test - Resistance Method	IEC 60034-1: 2010 cl. 8.4	Yes	Temp. rise(°C)	Max.	70
3	Load test & Efficiency Determination	IEC 60034-1: 2010 cl.9.4	Yes	η at 100% o/p	Min.	77.46
				Power Factor	Min.	0.71
				slip(%)	Min.	3.573
					Max.	5.360
4	Momentary Excess Torque test	IEC 60034-1: 2010 cl.9.4	Yes	Visual	-	-
5	Locked rotor test at reduced voltage	IEC 60034-1: 2010 cl.9.4	Yes	-	-	-
6	Locked rotor torque test	IS 12615: 2011 cl. 19.5	Yes	Ts/Tn	Min.	2.38
					Max	3.5
				Ist/In	Max.	6.96
7	Reduced Voltage running up test	IS 12615: 2011 cl. 19.4	Yes	Voltage(V)	Max.	239.61
8	No load test	IEC 60034-1: 2010 cl. 9.1	Yes	-	-	-
9	Measurement of Vibration severity	IEC 60034-14:2003 cl. 7.3	Yes	Vibration(mm/s)	Max.	1.3
10	Overspeed test at 1.2 times rated Speed	IEC 60034-1:2010 cl. 9.7	Yes	Visual Check	-	-
11	Insulation resistance test	IEC 60204-1: 2005 cl. 18.3	Yes	Insu. Res.	Min.	1MOhm
12	Polarization Index Test	IEEE 43-2013 cl.12.2.2	-	-	-	
13	Shaft Voltage Test	IEC 60034-9:2007 cl.4 & 5	-	-	-	-
14	Resistance measurement of Accessories	-	-	-	-	-
15	High Voltage Test	IEC 60204-1: 2005 cl. 18.3	Yes	Insu. Res.	Min.	1MΩ
16	Noise Test at No Load	IEC 60034-9:2007	-	-	-	-

Note: **This certificate shall not be reproduced, except in full, without written permission from Siemens AMTC authorities**

-This machine was tested on Test Bench ATI-1

Witnessed by	Tested by	Checked by	Authorized by		Authorized by
Date:	A.W.Shaikh	Nitin Rasal	Dilip Satav	or	Pramod Nagare
	Engineer	Engineer	Technical Manager		Deputy Technical Manager

Siemens Ltd.
RC-IN PD LD MF KW
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Thane Belapur Road, WMOT
Thane 400601, India

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Test Results

Resistance Measurement of Stator winding

Stator Winding	Terminal	Resistance (Ω)	Temperatue (°C)
Cold	U1 – V1	21.79000	30.1
	V1-W1	21.81000	
	U1-W1	21.81000	
Hot	U1 – V1	24.88000	32.5

Resistance measurement of accessories

RTD(PT100) (Ω)					
BTD (Ω)					
PTC (Ω)					
ACH (Ω)			Other		

Temperature Rise Test(by Resistance method)

Voltage (V)	Current (A)	Power (kW)	Power Factor	Slip (%)	Coolant Temp. (°C)	Temperature Rise(°C) RTD	DE Bearing	NDE Bearing	Frame
415.5	1.7	0.9312	0.76	4.114	32.5		22		24.8

Temperature rise after **3.11 hours** **35.2 °C**

Load Test and Efficiency Determination (by Indirect Method)

Output power (%)	Frequency (Hz)	Voltage (V)	Line Current (A)	Input Power (kW)	Slip (%)	Speed (RPM)	Output Power (kW)	Power Factor	Efficiency(η) %
150.0	49.97	415.10	2.41	1.480	6.871	1396.1	1.022	0.85	69.05
125.0	49.97	415.50	2.02	1.185	5.275	1420.0	0.884	0.82	74.58
100.0	49.97	415.40	1.71	0.934	3.989	1439.3	0.730	0.76	78.18
75.0	49.97	415.40	1.46	0.702	2.846	1456.4	0.559	0.67	79.66
50.0	49.97	415.10	1.27	0.485	1.825	1471.7	0.377	0.53	77.91
25.0	49.97	415.50	1.13	0.271	0.853	1486.3	0.183	0.33	67.60

Locked Rotor Test at reduced voltage and Locked rotor torque test

Frequency(Hz)	Voltage(V)	Current(A)	Power(kW)	Power Factor	Torque(Nm)	Tst/Tn	Ist/In
50.00	89.70	1.70	0.16	0.62	-	-	-
50.00	415.00	10.18	-	-	15.10	3.02	5.99

Reduced Voltage Running up test

Motor runs at full speed in specified direction on no load at Voltage (V) 110.4

No Load Test

Freq(Hz)	Voltage(V)	Current(A)	Power(kW)	pf	Speed
50.00	415	1.11	0.09	0.11	1499.1

Overspeed Test for 2 mins

Freq(Hz)	Speed(RPM)	Withstood
60.00	1798.00	

Shaft Voltage Test

Test Voltage(V)	Current(A)	Shaft Voltage(mV)
-	-	-

High Voltage Test

Voltage (kV)	Time(s)	Withstood
1.5	60	

Insulation Resistance Test

Insulation Resistance(MΩ)	
Before HV	>1000
After HV	>1000

Vibration severity Test at no load

Vibration (mm/s)	Axial	Horizontal	Vertical
DE	0.7	0.8	0.7
NDE		0.7	0.6

Polarisation Index Test

	Polarisation Index
IR_{10min}/IR_{1min}	-

Noise Level at No load

Background(dB)	Sound Level(dB)

Momentary Excess Torque test

Overload Test(%)	Time(s)	
160.00	15	Withstood

Legends: Tsn/Tn- Ratio of Starting Torque to Rated Torque of Machine
Ist/In-Ratio of Starting Current to Rated Current of Machine

Remarks-

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